

L1 Honeycomb Compression Test

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Signed: _____,

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Abstract

Honeycombs are frequently used materials in aerospace industry for their high strength-to-weight ratio and energy absorption under compression. This experiment examined the out-of-plane compressive behavior of an aluminum honeycomb following ASTM D 7336/D 7336M-07. Using the Tinius Olsen 150ST testing machine, the specimen showed a linear response at first, followed by cell wall buckling and gradual crushing, and eventual densification. These results provided useful insights into the structural energy absorption of honeycomb cores.

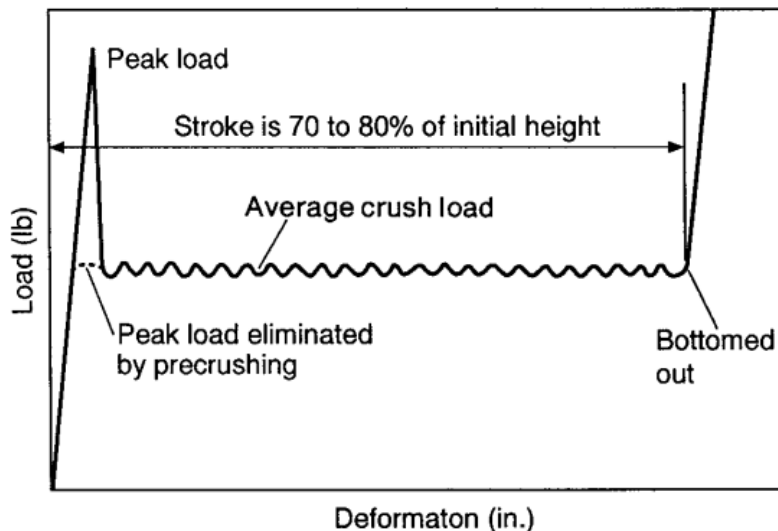
Introduction

Honeycomb structures are commonly used in the aerospace and other engineering fields because of their high strength to weight ratio, stiffness to weight ratio and good energy absorption. The shape of the honeycomb cores is designed to resist axial, bending and twisting loads, while being lightweight at the same time.

When loaded in compression, the thin cell walls bend buckle and collapse progressively which leads to 3 stages. First stage being the initial linear elastic region, second stage being the constant crush loading and third stage being the plastic deformation. The second region is the most important one as it demonstrates the high energy absorption of the honeycomb structure. ASTM D 7336/D 7336M – 07 [provides a standardized method for assessing this crushing behavior and defines parameters like the initial chord slope, average applied crush force, crush stroke and the usable energy absorbed during crushing.

The goal of the experiment was to examine and measure the out-of-plane compressive behavior of an aluminum honeycomb structure. The width, length, height, weight, cell size and foil thickness were measured before performing a compression test using a Tinius Olsen 150ST universal testing machine. The obtained raw data was then plotted into a Force vs Position and Stress vs Strain graphs that demonstrate the mechanical response of the honeycomb. On the other hand, microscopy demonstrated the cell wall buckling mechanism. The experiment data was then compared to the manufacturer reported data.

The graph below shows the expected behavior of the honeycomb specimen under loading.



T. Bitzer, *Honeycomb Technology*, Chapman & Hall, London (1997).

Equipment/Apparatus



Figure 1: Aluminum Honeycomb Specimen – Side View

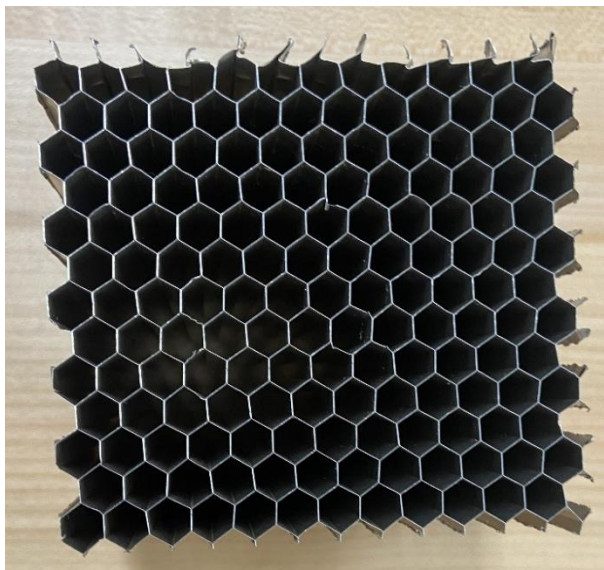


Figure 2: Aluminum Honeycomb Specimen – Top View



Figure 3: In-Plane Compressed Aluminum Honeycomb Specimen



Figure 4: Honeycomb Specimen during compression



Figure 5: Compressed Honeycomb Specimen

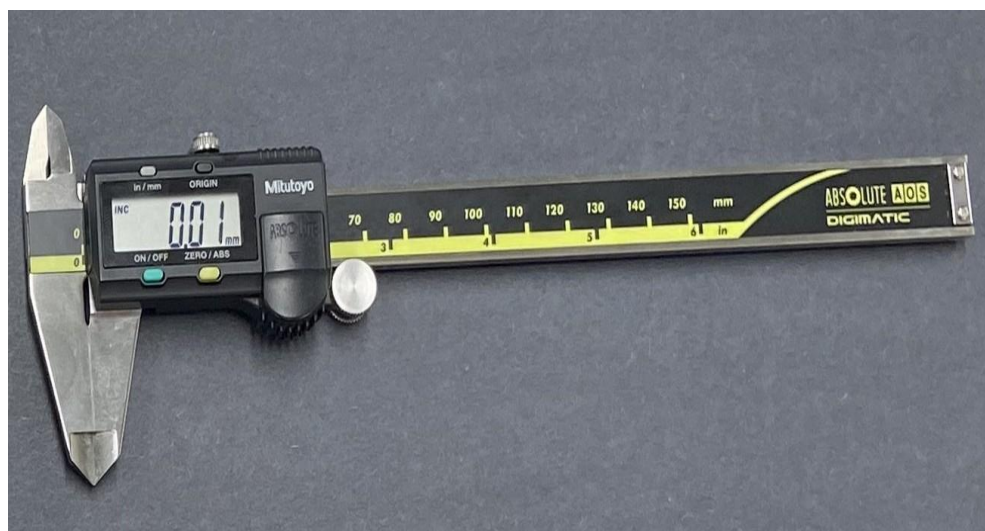


Figure 6: Calipers



Figure 7: Mass Scale



Figure 8: Tinius Olsen 150ST Electromechanical Universal Testing Machine

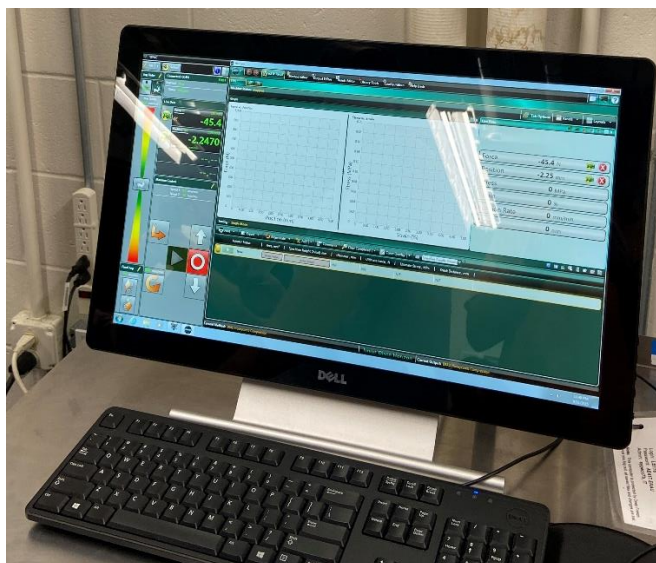


Figure 9: Computer and Monitor

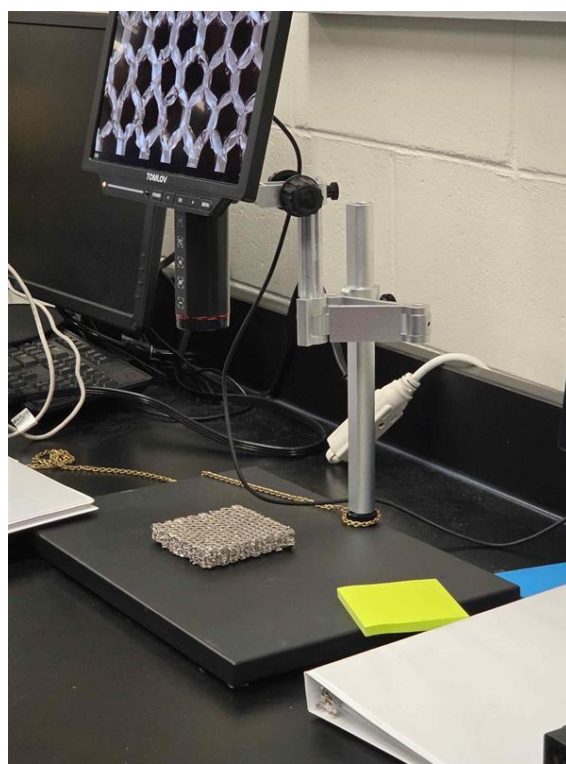


Figure 10: Motic stereo zoom microscope with illuminator, digital camera, WhiBal card, and imaging computer/software

Method

Four groups took four similar honeycombs. We took measurements of the cell size, length, width, height, and foil thickness of the specimen. An additional honeycomb was used to measure foil thickness, after it was crushed in-plane to flatten but not damage the cells.



Figure 11: Measuring the honeycomb's width.



Figure 12: Measuring the honeycomb's foil thickness.

The mean values of these were recorded in millimeters along with their standard deviation. The weight of the honeycomb was then taken, allowing the density to be calculated. The four honeycombs' means were then used to take a total mean; from this, the honeycomb that was both closest to the total mean and least damaged was selected for testing. The testing machine was then turned on with compression plates attached. The machine's method was changed to "ERAU Honeycomb Compression"; then the force was zeroed before any load was applied.

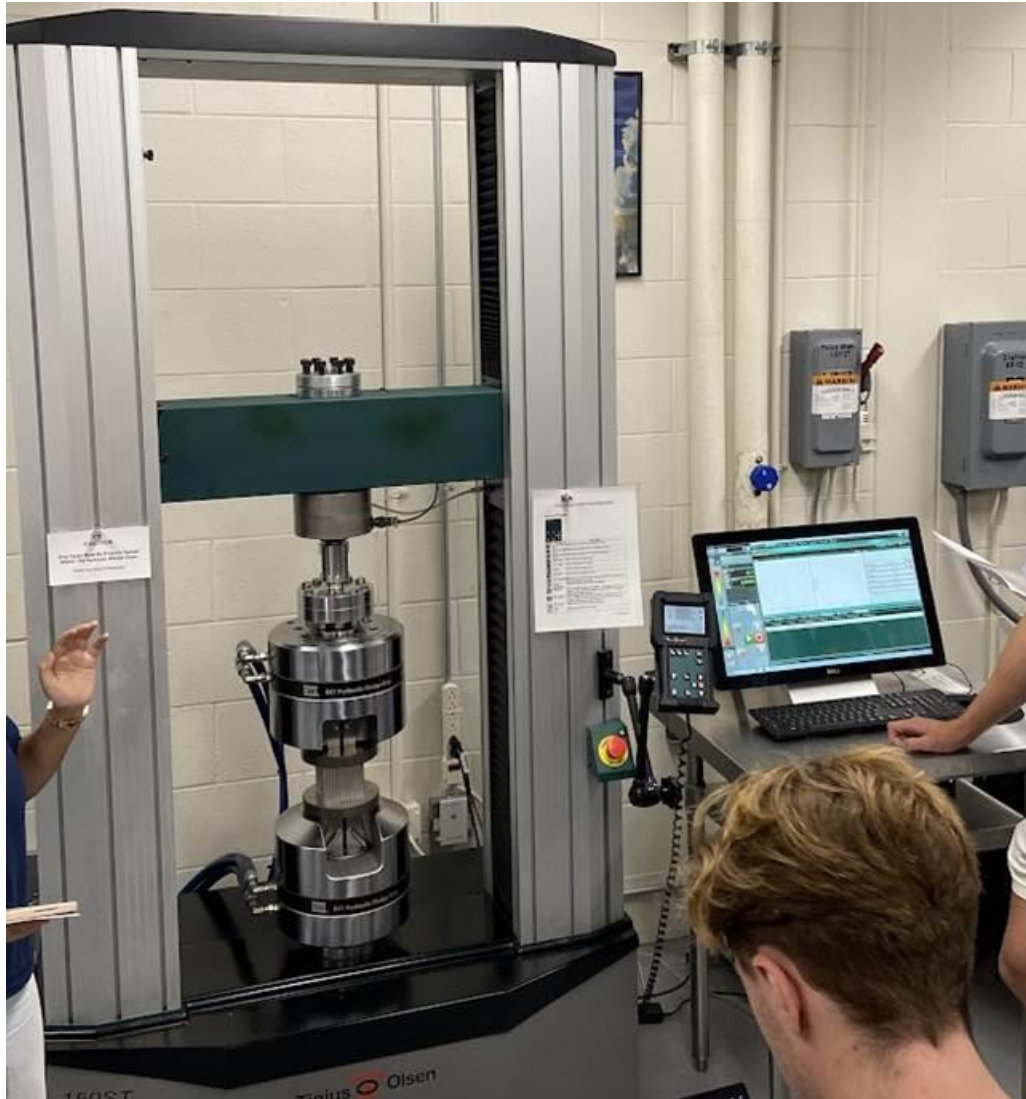


Figure 13: Honeycomb placed in the test machine pre-compression

The plates were moved to where the honeycomb could be placed in between. The plates were then moved to apply a pre-load of approximately -45 N on the honeycomb. The position was then zeroed. After entering the honeycomb's size data into the machine and attaching the safety shield, the machine was allowed to run its test and save the data.



Figure 14: Honeycomb at start of compression test



Figure 15: Honeycomb during compression test



Figure 16: Honeycomb post compression test

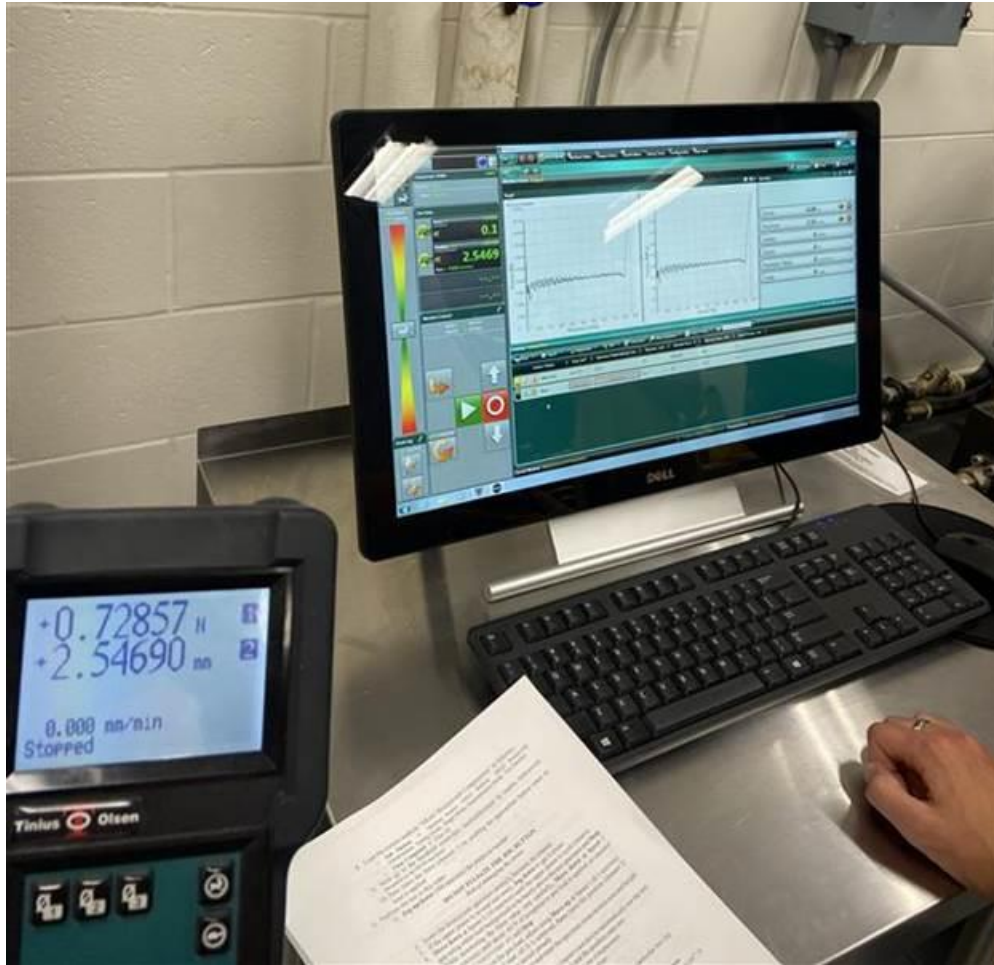


Figure 17: Computer data during compression test

After the data for the compression was gathered, we observed the crushed honeycomb's walls under a microscope.

Results

Table 1 shows the measurements recorded by our team, including the specimen weight, width, length, height, cell size and the foil thickness. Table 2 includes the mean measurements of all 4 teams, where the best quality honeycomb specimen is highlighted, which was used in the actual crushing experiment.

	1	2	3	4	5	Mean
Foil Thickness (mm)	0.078	0.074	0.077	0.067	0.066	0.0724
Length (mm)	78.0	77.5	77.5	73.0	78.0	76.8
Width (mm)	80.6	82.0	81.0	80.0	81.3	81.0
Height (mm)	75	75	75	75	75	75
Cell size (mm)	6.11	6.23	5.99	6.07	6.15	6.11
Mass (g)	36.25	36.25	36.24	36.24	36.24	36.244

Table 1: Our team's measurements

	Team 1	Team 2	Team 3	Team 4	Mean	Standard Deviation
Foil Thickness (mm)	0.0724	0.084	0.071	0.075	0.0756	0.0058
Length (mm)	76.8	78.69	79.81	77.74	78.26	1.29
Width (mm)	81.0	80.53	81.14	82.34	81.25	0.77
Height (mm)	75	76.17	76.10	75.87	75.79	0.54
Cell size (mm)	6.11	6.32	6.11	6.07	6.15	0.11
Mass (g)	36.244	35.90	36.39	36.01	36.14	0.22

Table 2: All teams' average measurements

Figure 18 shows the experiment data acquired from the Tinius Olsen 150ST machine. The graphs follow the expected mechanical response of a honeycomb structure, including the peak load and average crush load.

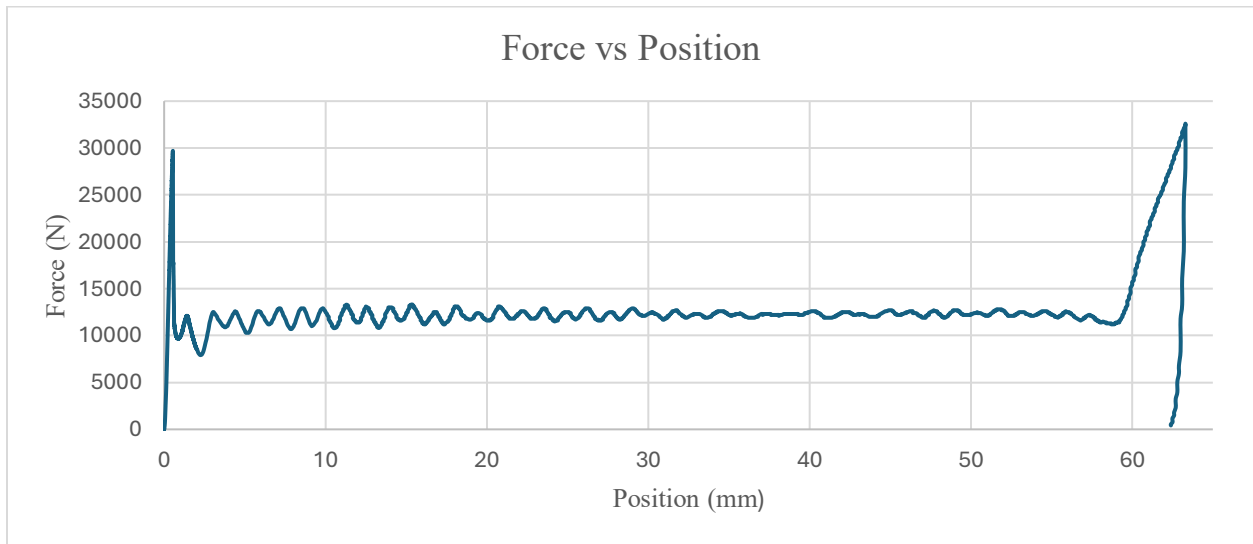


Figure 18: Force vs Position curve for honeycomb compression test

Figure 19 shows the stress-strain curve obtained from the experiment data.

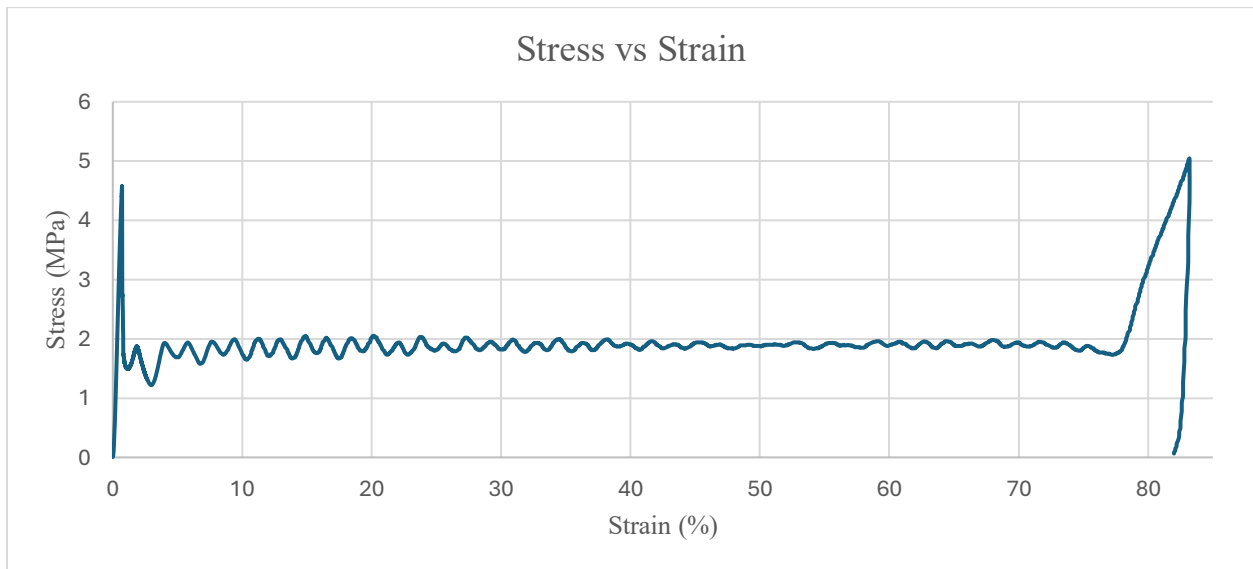


Figure 19: Stress vs Strain curve for honeycomb compression test

Figures 2020a-2020c show microscopy images of the aluminum honeycomb after the compression. The cell walls clearly show local buckling and folding, which corresponds to the figures above.

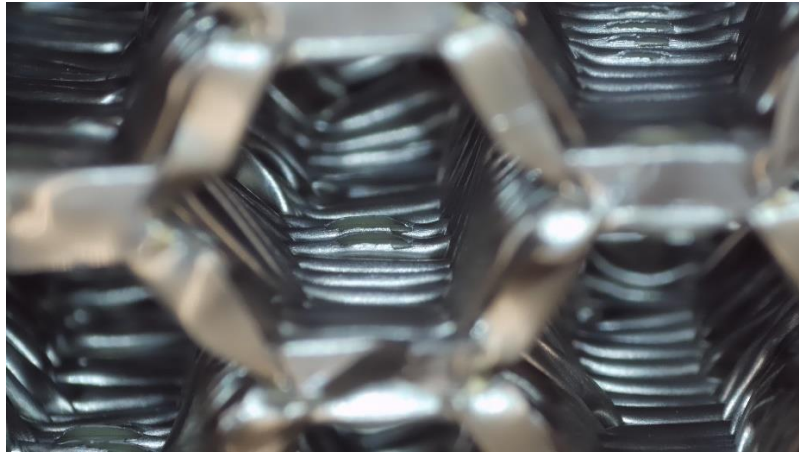


Figure 20a: Close-up view of bent and buckled honeycomb cell walls after compression

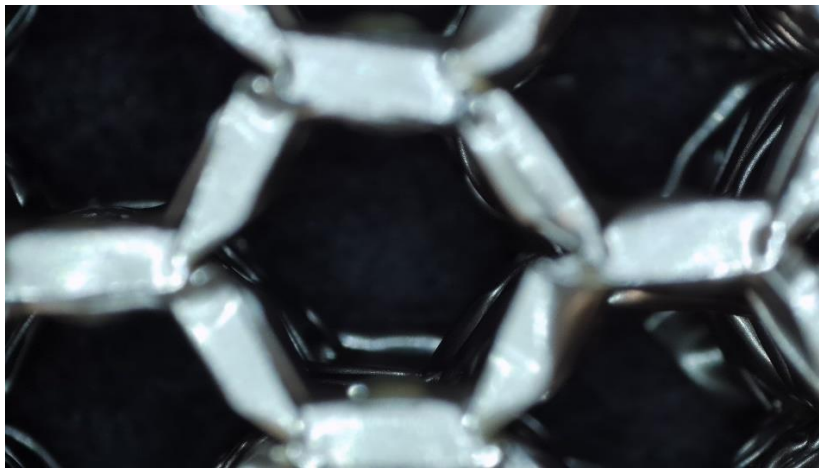


Figure 20b: Top view showing collapsed hexagonal geometry



Figure 20c: Deformation of adjacent cells showing wall bending

Analysis

F_{peak} , F_{crush} , σ_{peak} , and σ_{crush}

Looking at figure 18, we see that our F_{peak} and F_{crush} are 29300 N and 12000 N, respectively, with F_{peak} occurring at 0.535 mm, marking the loading peak before crushing begins and F_{crush} being the average force applied between F_{peak} and where the object is crushed passed its capabilities and bottoms out. F_{crush} being the left-over force after deformation occurs. Along with the forces, we have our σ_{crush} like F_{crush} , it is the average amount of stress still applied to the object after deformation occurs, averaging out to 1.87 MPa and σ_{peak} being 4.5 MPa.

Energy absorbed mathematically is simply the area under the curve, not including the area caused by $F_{\text{peak}}/\sigma_{\text{crush}}$, multiplied by the volume, which was calculated to be around 492.8 cm^3 using Team 3's data, it being the one decided on as the best fit. There are many ways to obtain this value, but we chose to break down the graph into two parts. As shown below, we can see that these two distinct sections are simply shapes with lengths and heights.

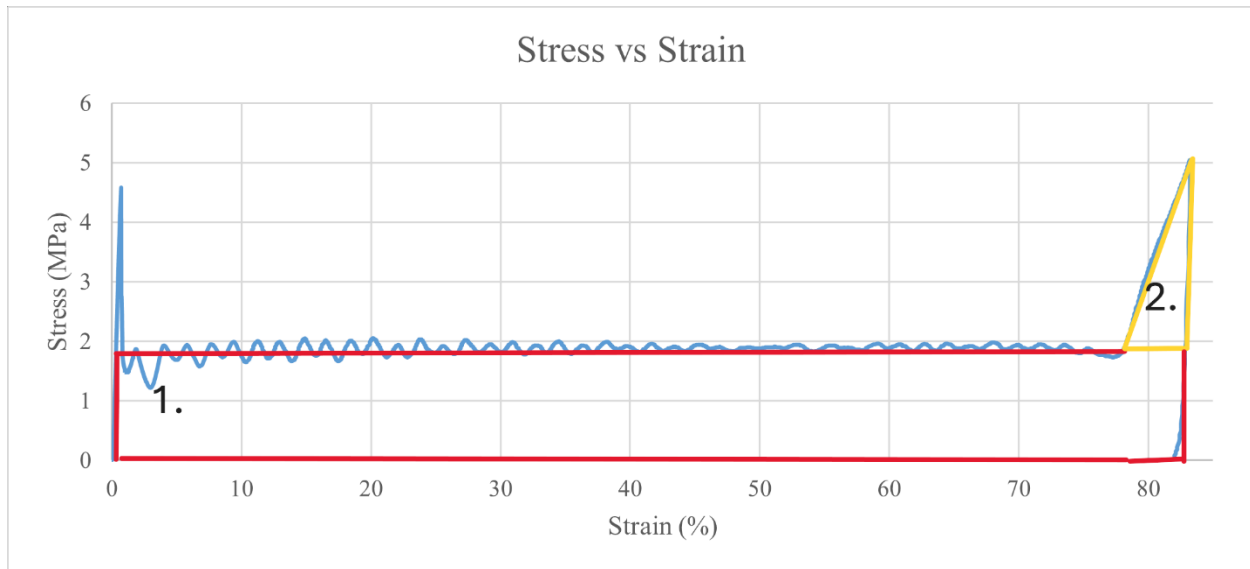


Figure 21: Altered Figure 19, Stress Vs. Strain Graph

Looking at the first area, we can see that the area here is simply σ_{crush} multiplied by the strain, which appears to be around 82.3%. Converting this to decimals gives us an area of around 1.54 J/mm^3 .

$$A_1 = H * L = \sigma_{\text{crush}} * \varepsilon = 1.87 \text{ MPa} * 0.823 = 1.54 \frac{\text{J}}{\text{mm}^3}$$

Applying this to area 2, we see that the area here is simply $\frac{1}{2}$ base times height. Starting with the height, the height of section 2 is the total height of both sections minus the height of section 1, or σ_{crush} .

$$A_2 = \frac{1}{2} * B * H = \frac{1}{2} * (\sigma_{total} - \sigma_{crush}) * H$$

Assuming we didn't have the original data, we can also estimate the total height to be a little over 5 MPa, subtracting this from σ_{crush} we get a height of 3.13 MPa. Using the raw data, we find that the total height is 5.05 MPa, giving section 2 a height of 3.18 MPa. We will continue using a height of 3.18 MPa. Now for the base, we can estimate it the same way we did for the height, giving us around 5.1% for the base. This gives us an area of 0.081 J/mm³.

$$A_2 = \frac{1}{2} * (5.05 - 1.87) MPa * 0.051 = 0.081 MPa$$

Finally, adding the two areas gives a total area of 1.621 J/mm³ and multiplying this by volume gives us the total energy absorbed, 838.834 [MPa*cm³]

$$E = V(A_1 + A_2) = 492.807 \text{ cm}^3 * (1.54 + 0.081) = 798.884 \text{ MPa} * \text{cm}^3$$

Starting with σ_{crush} and σ_{peak} , we have 1.87 MPa and 4.5 MPa respectively, equating to 270 psi and 653 psi. Along with the cell thickness being 0.071 mm equating to 0.0028 inches. Also calculating density using our volume from earlier along with the mass provided by team 3, we get a density of 4.6 pcf, and finally a cell size of 6.11 mm which comes out to a little under a quarter of an inch. The honeycomb type is most likely the 5056 Alloy Hexagonal Aluminum

5056 Alloy Hexagonal Aluminum Honeycomb – Specification Grade

Both CR-PAA and CR III corrosion-resistant coating

Hexcel Honeycomb Designation Cell Size – Alloy – Foil Gauge	Nominal Density pcf	Compressive					Crush Strength psi	Plate Shear					
		Bare		Stabilized				L Direction			W Direction		
		Strength psi		Strength psi		Modulus ksi		Strength psi		Modulus ksi	Strength psi		Modulus ksi
		typ	min	typ	min	typ	typ	typ	min	typ	typ	min	typ
1/16 – 5056 – .001	9.2	1700p	1300p	1800p	1400p	500p	850x	980p	760p	155.0p	600p	460p	50.0p
1/8 – 5056 – .0007	3.1	320	250	350	260	97	170	250	200	45.0	155	110	20.0
1/8 – 5056 – .001	4.5	630	475	690	500	185	320	440	350	70.0	255	205	28.0
1/8 – 5056 – .0015	6.1	1120	760	1200	825	295	535	690	525	102.0	400	305	38.0
1/8 – 5056 – .002	8.1	1750	1200	1900	1300	435	810	945	740	143.0	560	440	51.0
5/32 – 5056 – .0007	2.6	250	180	265	185	70	120	200	152	37.0	115	80	17.0
5/32 – 5056 – .001	3.8	450	360	500	375	140	235	335	272	57.0	195	155	24.0
5/32 – 5056 – .0015	5.3	820	615	865	650	240	420	550	435	85.0	325	250	33.0
5/32 – 5056 – .002	6.9	1220	920	1340	1000	350	650	760	610	118.0	430	360	43.0
3/16 – 5056 – .0007	2.0	190	110	200	120	45	75	140	105	27.0	85	50	13.0
3/16 – 5056 – .001	3.1	380	250	410	260	97	170	265	200	45.0	150	110	20.0
3/16 – 5056 – .0015	4.4	620	460	670	490	180	310	425	340	68.0	245	198	27.0
3/16 – 5056 – .002	5.7	920	685	1000	735	270	480	565	480	94.0	330	280	36.0
1/4 – 5056 – .0007	1.6	100	75	110	80	30	50	90	78	20.0	60	38	10.5
1/4 – 5056 – .001	2.3	240	145	265	155	58	100	180	130	32.0	100	62	15.0
1/4 – 5056 – .0015	3.4	400	300	480	315	115	200	290	230	50.0	175	130	22.0
1/4 – 5056 – .002	4.3	580	440	620	465	172	300	400	325	67.0	230	190	27.0

Figure 22: Hexcel Hexweb Honeycomb Attributes and Properties pg.

Honeycomb	Cell Size (in.)	σ_{crush} (psi)	σ_{peak} (psi)	Density (pcf)	Thickness (in.)
Target	0.25	300	620	4.3	0.002
Test	0.24	270	653	4.6	0.028

Table 3: Honeycomb Type Comparison of Test honeycomb vs the target honeycomb

Conclusion

The testing has concluded that the honeycomb is most likely a 5056 Alloy Hexagonal Honeycomb designation ¼-5056-0.002. With these, the test piece's values were not exact to the standard; this may be from minor errors in manufacturing or slight damage that the piece sustained before testing. During the test, the σ_{peak} was found to be at 4.5 MPA or 653 psi, the σ_{crush} was found to be 1.87 MPA or 270 psi, and the energy absorbed by the honeycomb was 798.884 MPa * cm³. These findings show the high out-of-plane compressive strength that honeycomb offers. When examining the compressed honeycomb under a microscope, this strength becomes visible as the walls buckle against each other. Even though one part of the cell is failing, the failed section will strengthen the next section that would buckle by acting as a buffer. In this buckling phase, the force varies only slightly. It is not decreasing like a non-honeycomb metal structure, and the energy absorbed during the process reflects this.

Acknowledgements

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A. Bharadwaj: Abstract and Equipment

J. Nguyen Jr.: Discussion/Analysis

R. Orr: Method and Conclusion

K. Satvaldy: Introduction and Results

References

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